MILITARY SPECIFICATION

PRIMER, ELECTRIC, M83 (ALTERNATE)
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-P-63311 (AR) dated 29 June 1979.

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Delete Paragraphs 4.5.6, 4.5.6.1, 4.5.6.2, 4.5.6.3 and 4.5.6.4 in their entirety and substitute the following:

"4.5.6 Determination of ignition charge composition.-The chemical composition of the ignition charge shall be considered satisfactory if it is in accordance with the following:

Component	Percentage
Potassium Chlorate & Gum Solids	46.1 ± 2.1%
Lead Thiocyanate	35.5 ± 2%
Charcoal	20.0 ± 1.5%

4.5.6.1 Charcoal.-Transfer approximately 2 grams of the igniter charge composition into a tared medium porosity crucible. Place in oven 70°C overnight to drive off water. Place crucible in desiccator to cool and then weigh to establish sample weight. Wash with ten 25 ml portion of 5 percent potassium nitrate solution followed by 50 ml of distilled water. Use a small glass rod to break up any lumps in the sample during the extraction process. Preserve the filtrate for the determination for lead thiocyanate. Place in oven maintained at 100 ± 5°C for 2 hours, cool in a desiccator & weigh. Calculate the percentage of charcoal as follows:

$$\text{Percent charcoal} = \frac{(A-B)(100)}{A}$$

A = Beginning sample weight
B = Empty crucible weight

4.5.6.2 Lead thiocyanate.-Exactly 35 ml of standardized 0.1 normal (N) silver nitrate shall be added to the filtrate reserved from the charcoal determination. The flask and contents shall be swirled and 100 ml of 3N nitric acid (nitric acid used should be water white and free of oxides of nitrogen) shall be added. The swirling of the mixture shall be continued for five minutes, 5 ml of a saturated solution of ferric ammonium sulfate be added and the solution titrated with standardized 0.1N potassium thiocyanate solution to the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follows:

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$$\text{Percent lead thiocyanate} = \frac{16.17\ (AB-CD)}{E}$$

A = Volume of silver nitrate solution used
B = Normality of silver nitrate solution
C = Volume of potassium thiocyanate solution
D = Normality of potassium thiocyanate
E = Weight of sample

4.5.6.3 Potassium chlorate & gum solids.-The percentage of potassium chlorate & gum solids is obtained by subtracting the total percentage of charcoal and lead thiocyanate from one hundred percent."

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